OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

STATION:	SALEM
SYSTEM:	Component Cooling Water (CCW)
TASK:	Perform Actions for CCW Restoration (23 CCW pump)
TASK NUMBER:	N1150420501
JPM NUMBER:	13-01 NRC Sim h

ALTERNATE PATH: [X] **K/A NUMBER:** 008 A4.01

IMPORTANCE FACTOR: 3.3 (**RO**) 3.1 (**SRO**)

APPLICABILITY:
EO [ ] **RO** [X] **STA** [ ] **SRO** [X]

EVALUATION SETTING/METHOD: Simulator

REFERENCES: 2-EOP-APPX-1 Rev. 24 (rev checked 9-6-14)

TOOLS AND EQUIPMENT: None

VALIDATED JPM COMPLETION TIME: 10 min

TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS: N/A

Developed By:	**G Gauding** **Instructor**	**Date: 9-6-14**
Validated By:	**C Omlor** **SME or Instructor**	**Date: 10-15-14**
Approved By:	Gregory Carney **Training Department**	**Date:** 10-23-14
Approved By:	(MOD) **Operations Department**	**Date:** 10-23-14

ACTUAL JPM COMPLETION TIME:

ACTUAL TIME CRITICAL COMPLETION TIME:

PERFORMED BY: ______________________

GRADE: [ ] **SAT** [ ] **UNSAT**

REASON, IF UNSATISFACTORY:

EVALUATOR'S SIGNATURE: **DATE:**

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NAME: ______________________

DATE: ______________________

SYSTEM: Component Cooling Water (CCW)

TASK: Perform Actions for CCW Restoration

TASK NUMBER: N1150420501

SIMULATOR SETUP: IC-258, markup 2-EOP-TRIP-1 through Step 17 direction to start one CCW pump IAW APPX-1. 22 CCW pump start PB O/R off.

INITIAL CONDITIONS:

The Unit 2 reactor was manually tripped and a SI initiated based on indications of a large LOCA. 2-EOP-TRIP-1 is in effect.

When the Main Generator breakers opened, all off site power was lost.

All vital buses are being powered from their respective EDGs.

NO CCW pumps are running.

INITIATING CUE:

You have been directed to start ONE CCW pump IAW 2-EOP-APPX-1.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:
1. Start 21 CCW pump IAW APPX-1.

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NAME: ______________________

DATE: ______________________

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TASK: Perform Actions for CCW Restoration

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Operator obtains 2-EOP-APPX-1			
	1	CHECK 4 KV vital bus status: a. CHECK AT LEAST ONE vital bus ENERGIZED by station power transformers.	Checks 4KV vital busses and determines no vital bus is being supplied from station power transformer and goes to Step 2.		
	2	CHECK ECCS pump (CVC;SI, or RHR) and AFW pump status: a. CHECK all ECCS pumps AND motor driven AFW pumps running on energized vital buses.	Checks running pumps and determines that ALL ECCS and motor driven AFW pumps are running on energized vital buses.		
	2.b.	CHECK one CCW pump running.	Checks CCW pump status and determines NO CCW pumps are running. Uses the RNO and goes to Step 3.		

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SYSTEM: Component Cooling Water (CCW)

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	3	SELECT CCW pump start strategy: a. IF ALL 4 KV vital buses energized, THEN GO TO Step 4.	 Determines ALL 4 KV vital buses energized, and GOES TO Step 4.		
	4	Start 22 CCW pump as follows:			
	4.a	CHECK 22 CCW pump AVAILABLE	Determines 22 CCW pump is AVAILABLE.		
*	4.b	BLOCK 2B and 2C SECs.	Blocks 2B and 2C SECs on 2RP1.		
*	4.c	RESET 2B and 2C SEC's	Resets 2B and 2C SEC's on 2CC3.		
	4.d	Perform the following at RP2:			
*	4.d.1	STOP 22 Switchgear Room Supply Fan.	Stops 22 Switchgear Room Supply Fan at RP2		
*	4.d.2	START 23 Switchgear Room Supply Fan.	Starts 23 Switchgear Room Supply Fan at RP2		

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JOB PERFORMANCE MEASURE

NAME: ____________________

DATE: ____________________

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TASK: Perform Actions for CCW Restoration

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	4.e	STOP the following equipment at CC1: • 22 CFCU • 24 CFCU • 22 Aux Bldg Exhaust Fan	Stops equipment at CC1: • 22 CFCU • 24 CFCU • 22 Aux Bldg Exhaust Fan		
*	4.f	START 22 CCW Pump.	Depresses start PB for 22 CCW Pump and recognizes it did not start.		
*	4.f RNO	1)START 22 OR 24 CFCU 2)GO TO Step 5	Starts 22 or 24 CFCU Goes to Step 5		
	5	START 21 CCW Pump as follows:			
*	5.a	BLOCK 2A AND 2B SECs.	Blocks 2 A SEC and verifies 2B SEC previously blocked.		
*	5.b	RESET 2A AND 2B SECs.	Resets 2A SEC and verifies 2B SEC previously reset.		
	5.c	SEND an operator to 100 ft elev chiller area to lock out 21 Chiller by placing the keyswitch in OFF (LOCKOUT) position.	Sends an operator to 100 ft elev chiller area to lock out 21 Chiller by placing the keyswitch in OFF (LOCKOUT) position.		
*	5.d	PERFORM the following at RP2: 1) STOP 21 Switchgear Room Supply Fan. 2) START 22 Switchgear Room Supply Fan.	At RP2, Stops 21 Switchgear Room Supply Fan and starts 22 Switchgear Room Supply Fan.		

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*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.e	STOP the following equipment at CC1: • 21 CFCU • 21 Aux Bldg Exhaust Fan	At CC1, stops 21 CFCU and 21 Aux Bldg Exhaust Fan		
*	5.f	START the following equipment at CC1: • 22 OR 24 CFCU • 22 FHB Exhaust Fan	At CC1 starts 22 or 24 CFCU and starts 22 FHB Exhaust Fan		
*	5.g	START 21 CCW Pump.	Starts 21 CCW pump.		
	6	PLACE 21 AND 22 CCW Heat Exchangers in service as follows:			
	6.a	CHECK AT LEAST THREE SW pumps running.	Determines 3 SW pumps are running.		
	6.b	SEND an operator to 84 ft elev Aux Bldg to place 21 AND 22 CCW Heat Exchangers in service IAW S2.OP-SO.CC-0002(Q), “21 & 22 Component Cooling Heat Exchanger Operation.”	Sends an operator to 84 ft elev Aux Bldg to place 21 AND 22 CCW Heat Exchangers in service IAW S2.OP-SO.CC-0002(Q), “21 & 22 Component Cooling Heat Exchanger Operation.”		
	6.c	RETURN TO procedure in effect.	Returns to procedure in effect.		

Terminating Cue: When operator announces returning to procedure in effect, state JPM is complete.

TQ-AA-106-0303

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

___ 1. Task description and number, JPM description and number are identified.

___ 2. Knowledge and Abilities (K/A) references are included.

___ 3. Performance location specified. (in-plant, control room, or simulator)

___ 4. Initial setup conditions are identified.

___ 5. Initiating and terminating Cues are properly identified.

___ 6. Task standards identified and verified by SME review.

___ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).

___ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev 24 Date 10-15-14

___ 9. Pilot test the JPM:
- a. verify Cues both verbal and visual are free of conflict, and
- b. ensure performance time is accurate.

___ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.

___ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: OMCar Date: 10-15-14

SME/Instructor: ____________ Date: ________

SME/Instructor: ____________ Date: ________

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